



**SWAMINARAYAN
UNIVERSITY**
THE RESERVOIR OF ALL KNOWLEDGE

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CHAPTER -1

1. Symmetric Cipher Model

The **Symmetric Cipher Model** is a cornerstone of classical cryptography, where a **single secret key** is used for both encryption and decryption.

Symmetric Encryption is the most basic and old method of encryption. It uses only one key for the process of both the encryption and decryption of data. Thus, it is also known as Single-Key Encryption.

A few basic terms in Cryptography are as follows:

Plain Text: original message to be communicated between sender and receiver

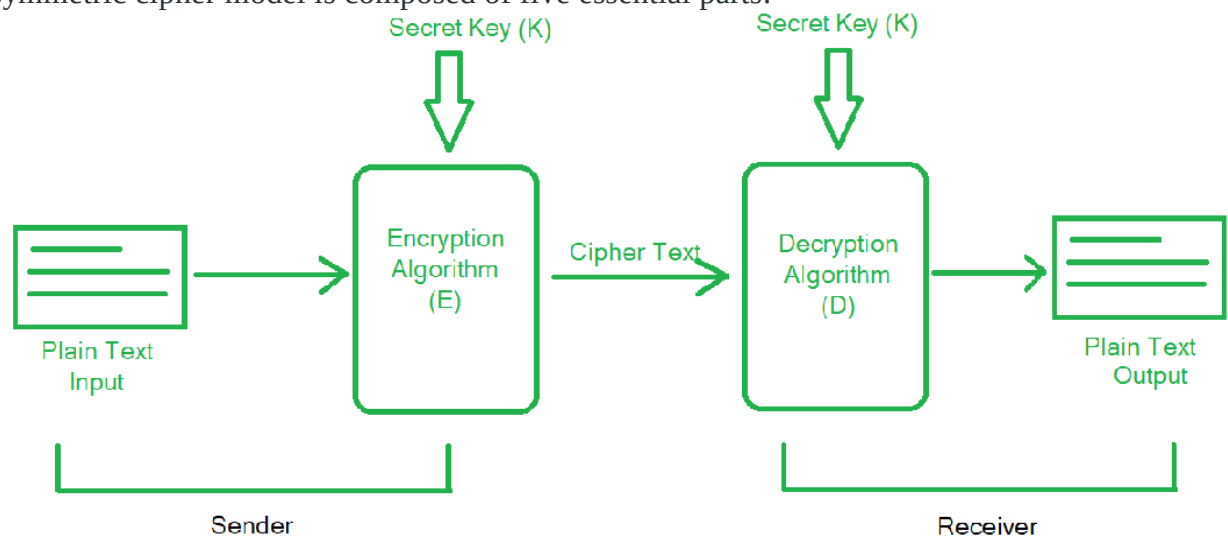
Cipher Text: encoded format of the original message that cannot be understood by humans

Encryption (or Enciphering): the conversion of plain text to cipher text

Decryption (or Deciphering): the conversion of cipher text to plain text, i.e., reverse of encryption

The Symmetric Cipher Model:

A symmetric cipher model is composed of five essential parts:



1. Plain Text (x): This is the original data/message that is to be communicated to the receiver by the sender. It is one of the inputs to the encryption algorithm.

2. Secret Key (k): It is a value/string/text file used by the encryption and decryption algorithm to encode and decode the plain text to cipher text and vice-versa respectively. It is independent of the encryption algorithm. It governs all the conversions in plain text. All the substitutions and transformations done depend on the secret key.

3. Encryption Algorithm (E): It takes the plain text and the secret key as inputs and produces Cipher Text as output. It implies several techniques such as substitutions and transformations on the plain text using the secret key.

$$E(x, k) = y$$

4. Cipher Text (y): It is the formatted form of the plain text (x) which is unreadable for humans, hence providing encryption during the transmission. It is completely dependent upon the secret key provided to the encryption algorithm. Each unique secret key produces a unique cipher text.

5. Decryption Algorithm (D): It performs reversal of the encryption algorithm at the recipient's side. It also takes the secret key as input and decodes the cipher text received from the sender based on the secret key. It produces plain text as output.

$$D(y, k) = x$$

Requirements for Encryption:

There are only two requirements that need to be met to perform encryption. They are,

1. Encryption Algorithm: There is a need for a very strong encryption algorithm that produces cipher texts in such a way that the attacker should be unable to crack the secret key even if they have access to one or more cipher texts.

2. Secure way to share Secret Key: There must be a secure and robust way to share the secret key between the sender and the receiver. It should be leak proof so that the attacker cannot access the secret key.

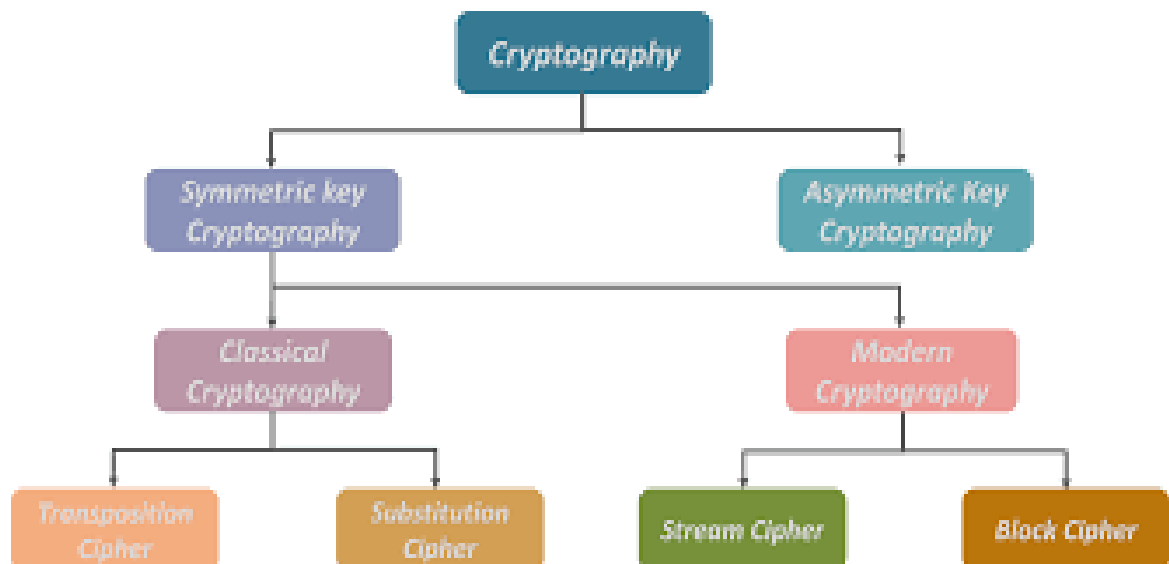
2. Cryptography

Cryptography: The Art of Secure Communication

Cryptography is the science and practice of securing information by transforming it into a format that is unreadable to unauthorized users. It ensures **confidentiality**, **integrity**, **authentication**, and **non-repudiation** of data.

Types of Cryptography

Type	Description
Symmetric Key	Same key used for encryption and decryption. Fast but requires secure key sharing.
Asymmetric Key	Uses a public key for encryption and a private key for decryption. More secure for key exchange.
Hash Functions	Converts data into a fixed-length hash. Used for integrity checks, not reversible.



Core Concepts

- **Plaintext:** The original readable data.
- **Cipher text:** The encrypted, unreadable version of the data.
- **Encryption:** The process of converting plaintext into cipher text.
- **Decryption:** Reversing cipher text back into plaintext.
- **Key:** A string of bits used by encryption algorithms to transform data.

3. Cryptanalysis and Attacks

Cryptanalysis is the art and science of deciphering encrypted data without access to the secret key. It's used to test the strength of cryptographic systems and, in some cases, to exploit their weaknesses.

Purpose of Cryptanalysis

- **Evaluate security** of encryption algorithms
- **Identify vulnerabilities** in cryptographic protocols
- **Recover plaintext or keys** from cipher text
- **Strengthen systems** by learning from weaknesses

Types of Cryptanalytic Attacks

Attack Type	Description
Cipher text-Only Attack (COA)	Attacker has only cipher text and tries to deduce the plaintext or key
Known-Plaintext Attack (KPA)	Attacker knows some plaintext-cipher text pairs and uses them to find the key
Chosen-Plaintext Attack (CPA)	Attacker chooses plaintexts and obtains cipher texts to analyse the algorithm
Adaptive Chosen-Plaintext Attack (ACPA)	Attacker refines plaintext choices based on previous results
Chosen-Cipher text Attack (CCA)	Attacker decrypts chosen cipher texts to learn about the key or algorithm
Man-in-the-Middle (MITM)	Attacker intercepts communication between two parties to steal or alter data
Side-Channel Attack	Exploits physical characteristics like timing, power usage, or EM emissions
Brute-Force Attack	Tries every possible key until the correct one is found

Attack Type	Description
Dictionary Attack	Uses a precompiled list of possible keys or passwords
Rainbow Table Attack	Uses precomputed hash values to reverse-engineer passwords
Differential Cryptanalysis	Analyses differences in input-output pairs to find patterns
Linear Cryptanalysis	Uses linear approximations to deduce key bits from plaintext-cipher text pairs

4. Substitution and Transposition techniques

These are two foundational methods used in classical encryption to transform plaintext into cipher text.

Substitution Techniques

Substitution replaces elements of the plaintext with another element—characters, symbols, or numbers.

- **Simple Substitution Cipher:** Each letter is replaced by another (e.g., Caesar cipher where A becomes D if shifted by 3).
- **Monoalphabetic Cipher:** Uses a single fixed substitution over the entire message.
- **Polyalphabetic Cipher:** Uses multiple substitution alphabets to make decryption harder (e.g., Vigenère cipher).
- **Homophonic Substitution:** Each plaintext letter can be substituted by several cipher text symbols, increasing complexity.

The goal is to disguise the actual symbols while keeping their order.

Transposition Techniques

Transposition changes the order of characters in the plaintext without altering the characters themselves.

- **Rail Fence Cipher:** Characters are written in a zigzag pattern across multiple “rails” and then read row-wise.
- **Columnar Transposition:** Text is written into columns based on a keyword, then columns are rearranged.
- **Permutation Cipher:** The plaintext is broken into blocks and the characters within each block are permuted using a fixed pattern.

Combined Use

Many cryptographic systems use both substitution and transposition together to strengthen the encryption process—for instance, the Data Encryption Standard (DES) uses both concepts.
